

12 V input / 100 W upstream USB-PD / 2 USB-C + 4 USB-A downstream
Engineering draft – not released for manufacture

Sheet: /
File: XG_Mobile_USB_Hub.kicad_sch

Title: XG Mobile USB Hub

Size: A3Date:KiCad E.D.A. 9.0.9

Rev: DRAFT AId: 1/4

USB–C DAUGHTERBOARD / ENGINEERING DRAFT

Upstream: USB–C, USB 3.2 Gen 2 data UFP, USB–PD power source, 20 V / 5 A maximum.
Downstream: 2 Type–C + 3 Type–A SuperSpeed, 1 Type–A USB 2.0.
Use two GPU power inputs OR one EPS input. Not for simultaneous independent supplies.

Power Supplies and Inputs

File: XG_Mobile_USB_Hub_Power.kicad_sch

MCU and Upstream PD Control

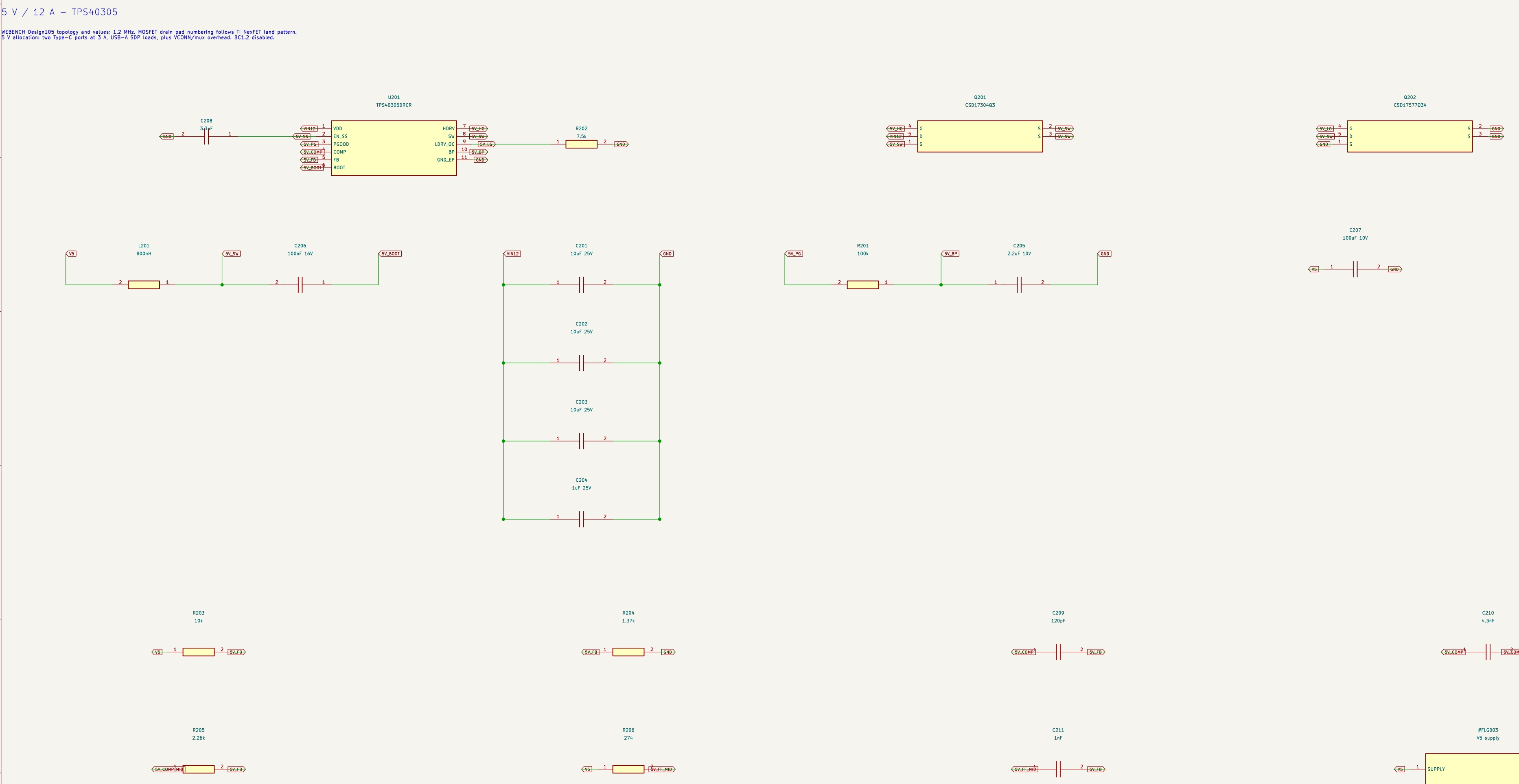
File: XG_Mobile_USB_Hub_MCU.kicad_sch

USB Hub and Downstream Ports

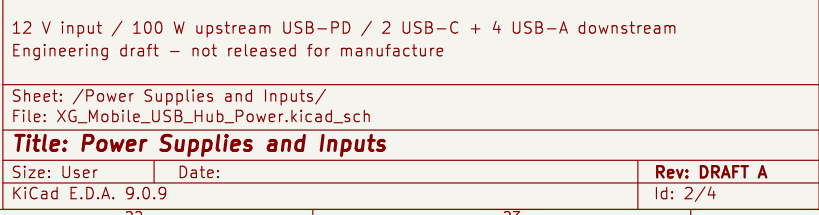
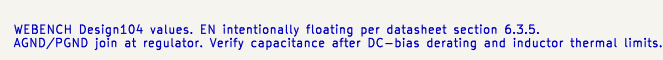
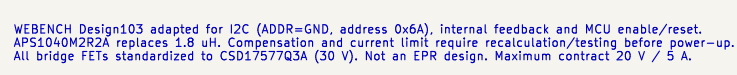
File: XG_Mobile_USB_Hub_USB_Hub.kicad_sch

Named nets connect the circuit blocks on this sheet and across the hierarchy.

Choose GPU pair @8 FPS with a break-before-make selector rated ≥ 30 A, ≥ 25 VDC. Do not switch under load. GPU pair H251 came from the same PSU. Both GPU connectors required for full output. Do not combine separate PSUs. Connector pin numbering is the electrical interface; verify keying, mating face and selected manufacturer drawing before layout.



TPS82133, 12 V input, 0.8 V reference $\times [1 + 63.7\%/100\%] = 1.3496$ V.
 Its V_{OSR1} allowed range 1.09–1.21 V. Hold H03_DSCT_N low until both bus rails are stable.



MCU and Upstream PD Control

Named nets connect the circuit blocks on this sheet and across the hierarchy.

STM32G071 – USB-PD policy and power control

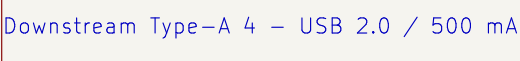
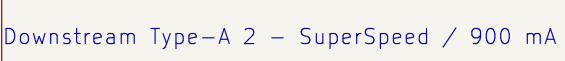
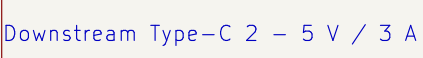
STM32G071x85M (QFP33) PA6-UCPD1_CCL, PA65-UCPD1_CCL, PA66-PB7-UC1, PA13/PA14-SD0.
Source=any, power=any, uCPD dead-battery data grounded, enable dead-battery function in firmware.
Reset from flash using option bytes, internal BIST function. Firmware required: no PD stack is generated here.

Upstream Type-C – USB data UFP / PD source

STM32G071x85M (QFP33) PA6-UCPD1_CCL, PA65-UCPD1_CCL, PA66-PB7-UC1, PA13/PA14-SD0.
TCPRX source path, control=source, data=any, 2.5V, 5V, 9V, 12V, 15V, 20V, 24V, 28V, 30V, 33V, 36V, 39V, 42V, 45V, 48V, 51V, 54V, 57V, 60V, 63V, 66V, 69V, 72V, 75V, 78V, 81V, 84V, 87V, 90V, 93V, 96V, 99V, 102V, 105V, 108V, 111V, 114V, 117V, 120V, 123V, 126V, 129V, 132V, 135V, 138V, 141V, 144V, 147V, 150V, 153V, 156V, 159V, 162V, 165V, 168V, 171V, 174V, 177V, 180V, 183V, 186V, 189V, 192V, 195V, 198V, 201V, 204V, 207V, 210V, 213V, 216V, 219V, 222V, 225V, 228V, 231V, 234V, 237V, 240V, 243V, 246V, 249V, 252V, 255V, 258V, 261V, 264V, 267V, 270V, 273V, 276V, 279V, 282V, 285V, 288V, 291V, 294V, 297V, 300V, 303V, 306V, 309V, 312V, 315V, 318V, 321V, 324V, 327V, 330V, 333V, 336V, 339V, 342V, 345V, 348V, 351V, 354V, 357V, 360V, 363V, 366V, 369V, 372V, 375V, 378V, 381V, 384V, 387V, 390V, 393V, 396V, 399V, 402V, 405V, 408V, 411V, 414V, 417V, 420V, 423V, 426V, 429V, 432V, 435V, 438V, 441V, 444V, 447V, 450V, 453V, 456V, 459V, 462V, 465V, 468V, 471V, 474V, 477V, 480V, 483V, 486V, 489V, 492V, 495V, 498V, 501V, 504V, 507V, 510V, 513V, 516V, 519V, 522V, 525V, 528V, 531V, 534V, 537V, 540V, 543V, 546V, 549V, 552V, 555V, 558V, 561V, 564V, 567V, 570V, 573V, 576V, 579V, 582V, 585V, 588V, 591V, 594V, 597V, 600V, 603V, 606V, 609V, 612V, 615V, 618V, 621V, 624V, 627V, 630V, 633V, 636V, 639V, 642V, 645V, 648V, 651V, 654V, 657V, 660V, 663V, 666V, 669V, 672V, 675V, 678V, 681V, 684V, 687V, 690V, 693V, 696V, 699V, 702V, 705V, 708V, 711V, 714V, 717V, 720V, 723V, 726V, 729V, 732V, 735V, 738V, 741V, 744V, 747V, 750V, 753V, 756V, 759V, 762V, 765V, 768V, 771V, 774V, 777V, 780V, 783V, 786V, 789V, 792V, 795V, 798V, 801V, 804V, 807V, 810V, 813V, 816V, 819V, 822V, 825V, 828V, 831V, 834V, 837V, 840V, 843V, 846V, 849V, 852V, 855V, 858V, 861V, 864V, 867V, 870V, 873V, 876V, 879V, 882V, 885V, 888V, 891V, 894V, 897V, 900V, 903V, 906V, 909V, 912V, 915V, 918V, 921V, 924V, 927V, 930V, 933V, 936V, 939V, 942V, 945V, 948V, 951V, 954V, 957V, 960V, 963V, 966V, 969V, 972V, 975V, 978V, 981V, 984V, 987V, 990V, 993V, 996V, 999V, 1002V, 1005V, 1008V, 1011V, 1014V, 1017V, 1020V, 1023V, 1026V, 1029V, 1032V, 1035V, 1038V, 1041V, 1044V, 1047V, 1050V, 1053V, 1056V, 1059V, 1062V, 1065V, 1068V, 1071V, 1074V, 1077V, 1080V, 1083V, 1086V, 1089V, 1092V, 1095V, 1098V, 1101V, 1104V, 1107V, 1110V, 1113V, 1116V, 1119V, 1122V, 1125V, 1128V, 1131V, 1134V, 1137V, 1140V, 1143V, 1146V, 1149V, 1152V, 1155V, 1158V, 1161V, 1164V, 1167V, 1170V, 1173V, 1176V, 1179V, 1182V, 1185V, 1188V, 1191V, 1194V, 1197V, 1200V, 1203V, 1206V, 1209V, 1212V, 1215V, 1218V, 1221V, 1224V, 1227V, 1230V, 1233V, 1236V, 1239V, 1242V, 1245V, 1248V, 1251V, 1254V, 1257V, 1260V, 1263V, 1266V, 1269V, 1272V, 1275V, 1278V, 1281V, 1284V, 1287V, 1290V, 1293V, 1296V, 1299V, 1302V, 1305V, 1308V, 1311V, 1314V, 1317V, 1320V, 1323V, 1326V, 1329V, 1332V, 1335V, 1338V, 1341V, 1344V, 1347V, 1350V, 1353V, 1356V, 1359V, 1362V, 1365V, 1368V, 1371V, 1374V, 1377V, 1380V, 1383V, 1386V, 1389V, 1392V, 1395V, 1398V, 1401V, 1404V, 1407V, 1410V, 1413V, 1416V, 1419V, 1422V, 1425V, 1428V, 1431V, 1434V, 1437V, 1440V, 1443V, 1446V, 1449V, 1452V, 1455V, 1458V, 1461V, 1464V, 1467V, 1470V, 1473V, 1476V, 1479V, 1482V, 1485V, 1488V, 1491V, 1494V, 1497V, 1500V, 1503V, 1506V, 1509V, 1512V, 1515V, 1518V, 1521V, 1524V, 1527V, 1530V, 1533V, 1536V, 1539V, 1542V, 1545V, 1548V, 1551V, 1554V, 1557V, 1560V, 1563V, 1566V, 1569V, 1572V, 1575V, 1578V, 1581V, 1584V, 1587V, 1590V, 1593V, 1596V, 1599V, 1602V, 1605V, 1608V, 1611V, 1614V, 1617V, 1620V, 1623V, 1626V, 1629V, 1632V, 1635V, 1638V, 1641V, 1644V, 1647V, 1650V, 1653V, 1656V, 1659V, 1662V, 1665V, 1668V, 1671V, 1674V, 1677V, 1680V, 1683V, 1686V, 1689V, 1692V, 1695V, 1698V, 1701V, 1704V, 1707V, 1710V, 1713V, 1716V, 1719V, 1722V, 1725V, 1728V, 1731V, 1734V, 1737V, 1740V, 1743V, 1746V, 1749V, 1752V, 1755V, 1758V, 1761V, 1764V, 1767V, 1770V, 1773V, 1776V, 1779V, 1782V, 1785V, 1788V, 1791V, 1794V, 1797V, 1800V, 1803V, 1806V, 1809V, 1812V, 1815V, 1818V, 1821V, 1824V, 1827V, 1830V, 1833V, 1836V, 1839V, 1842V, 1845V, 1848V, 1851V, 1854V, 1857V, 1860V, 1863V, 1866V, 1869V, 1872V, 1875V, 1878V, 1881V, 1884V, 1887V, 1890V, 1893V, 1896V, 1899V, 1902V, 1905V, 1908V, 1911V, 1914V, 1917V, 1920V, 1923V, 1926V, 1929V, 1932V, 1935V, 1938V, 1941V, 1944V, 1947V, 1950V, 1953V, 1956V, 1959V, 1962V, 1965V, 1968V, 1971V, 1974V, 1977V, 1980V, 1983V, 1986V, 1989V, 1992V, 1995V, 1998V, 2001V, 2004V, 2007V, 2010V, 2013V, 2016V, 2019V, 2022V, 2025V, 2028V, 2031V, 2034V, 2037V, 2040V, 2043V, 2046V, 2049V, 2052V, 2055V, 2058V, 2061V, 2064V, 2067V, 2070V, 2073V, 2076V, 2079V, 2082V, 2085V, 2088V, 2091V, 2094V, 2097V, 2100V, 2103V, 2106V, 2109V, 2112V, 2115V, 2118V, 2121V, 2124V, 2127V, 2130V, 2133V, 2136V, 2139V, 2142V, 2145V, 2148V, 2151V, 2154V, 2157V, 2160V, 2163V, 2166V, 2169V, 2172V, 2175V, 2178V, 2181V, 2184V, 2187V, 2190V, 2193V, 2196V, 2199V, 2202V, 2205V, 2208V, 2211V, 2214V, 2217V, 2220V, 2223V, 2226V, 2229V, 2232V, 2235V, 2238V, 2241V, 2244V, 2247V, 2250V, 2253V, 2256V, 2259V, 2262V, 2265V, 2268V, 2271V, 2274V, 2277V, 2280V, 2283V, 2286V, 2289V, 2292V, 2295V, 2298V, 2301V, 2304V, 2307V, 2310V, 2313V, 2316V, 2319V, 2322V, 2325V, 2328V, 2331V, 2334V, 2337V, 2340V, 2343V, 2346V, 2349V, 2352V, 2355V, 2358V, 2361V, 2364V, 2367V, 2370V, 2373V, 2376V, 2379V, 2382V, 2385V, 2388V, 2391V, 2394V, 2397V, 2400V, 2403V, 2406V, 2409V, 2412V, 2415V, 2418V, 2421V, 2424V, 2427V, 2430V, 2433V, 2436V, 2439V, 2442V, 2445V, 2448V, 2451V, 2454V, 2457V, 2460V, 2463V, 2466V, 2469V, 2472V, 2475V, 2478V, 2481V, 2484V, 2487V, 2490V, 2493V, 2496V, 2499V, 2502V, 2505V, 2508V, 2511V, 2514V, 2517V, 2520V, 2523V, 2526V, 2529V, 2532V, 2535V, 2538V, 2541V, 2544V, 2547V, 2550V, 2553V, 2556V, 2559V, 2562V, 2565V, 2568V, 2571V, 2574V, 2577V, 2580V, 2583V, 2586V, 2589V, 2592V, 2595V, 2598V, 2601V, 2604V, 2607V, 2610V, 2613V, 2616V, 2619V, 2622V, 2625V, 2628V, 2631V, 2634V, 2637V, 2640V, 2643V, 2646V, 2649V, 2652V, 2655V, 2658V, 2661V, 2664V, 2667V, 2670V, 2673V, 2676V, 2679V, 2682V, 2685V, 2688V, 2691V, 2694V, 2697V, 2700V, 2703V, 2706V, 2709V, 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3141V, 3144V, 3147V, 3150V, 3153V, 3156V, 3159V, 3162V, 3165V, 3168V, 3171V, 3174V, 3177V, 3180V, 3183V, 3186V, 3189V, 3192V, 3195V, 3198V, 3201V, 3204V, 3207V, 3210V, 3213V, 3216V, 3219V, 3222V, 3225V, 3228V, 3231V, 3234V, 3237V, 3240V, 3243V, 3246V, 3249V, 3252V, 3255V, 3258V, 3261V, 3264V, 3267V, 3270V, 3273V, 3276V, 3279V, 3282V, 3285V, 3288V, 3291V, 3294V, 3297V, 3300V, 3303V, 3306V, 3309V, 3312V, 3315V, 3318V, 3321V, 3324V, 3327V, 3330V, 3333V, 3336V, 3339V, 3342V, 3345V, 3348V, 3351V, 3354V, 3357V, 3360V, 3363V, 3366V, 3369V, 3372V, 3375V, 3378V, 3381V, 3384V, 3387V, 3390V, 3393V, 3396V, 3399V, 3402V, 3405V, 3408V, 3411V, 3414V, 3417V, 3420V, 3423V, 3426V, 3429V, 3432V, 3435V, 3438V, 3441V, 3444V, 3447V, 3450V, 3453V, 3456V, 3459V, 3462V, 3465V, 3468V, 3471V, 3474V, 3477V, 3480V, 3483V, 3486V, 3489V, 3492V, 3495V, 3498V, 3501V, 3504V, 3507V, 3510V, 3513V, 3516V, 3519V, 3522V, 3525V, 3528V, 3531V, 3534V, 3537V, 3540V, 3543V, 3546V, 3549V, 3552V, 3555V, 3558V, 3561V, 3564V, 3567V, 3570V, 3573V, 3576V, 3579V, 3582V, 3585V, 3588V, 3591V, 3594V, 3597V, 3600V, 3603V, 3606V, 3609V, 3612V, 3615V, 3618V, 3621V, 3624V, 3627V, 3630V, 3633V, 3636V, 3639V, 3642V, 3645V, 3648V, 3651V, 3654V, 3657V, 3660V, 3663V, 3666V, 3669V, 3672V, 3675V, 3678V, 3681V, 3684V, 3687V, 3690V, 3693V, 3696V, 3699V, 3702V, 3705V, 3708V, 3711V, 3714V, 3717V, 3720V, 3723V, 3726V, 3729V, 3732V, 3735V, 3738V, 3741V, 3744V, 3747V, 3750V, 3753V, 3756V, 3759V, 3762V, 3765V, 3768V, 3771V, 3774V, 3777V, 3780V, 3783V, 3786V, 3789V, 3792V, 3795V, 3798V, 3801V, 3804V, 3807V, 3810V, 3813V, 3816V, 3819V, 3822V, 3825V, 3828V, 3831V, 3834V, 3837V, 3840V, 3843V, 3846V, 3849V, 3852V, 3855V, 3858V, 3861V, 3864V, 3867V, 3870V, 3873V, 3876V, 3879V, 3882V, 3885V, 3888V, 3891V, 3894V, 3897V, 3900V, 3903V, 3906V, 3909V, 3912V, 3915V, 3918V, 3921V, 3924V, 3927V, 3930V, 3933V, 3936V, 3939V, 3942V, 3945V, 3948V, 3951V, 3954V, 3957V, 3960V, 3963V, 3966V, 3969V, 3972V, 3975V, 3978V, 3981V, 3984V, 3987V, 3990V, 3993V, 3996V, 3999V, 4002V, 4005V, 4008V, 4011V, 4014V, 4017V, 4020V, 4023V, 4026V, 4029V, 4032V, 4035V, 4038V, 4041V, 4044V, 4047V, 4050V, 4053V, 4056V, 4059V, 4062V, 4065V, 4068V, 4071V, 4074V, 4077V, 4080V, 4083V, 4086V, 4089V, 4092V, 4095V, 4098V, 4101V, 4104V, 4107V, 4110V, 4113V, 4116V, 4119V, 4122V, 4125V, 4128V, 4131V, 4134V, 4137V, 4140V, 4143V, 4146V, 4149V, 4152V, 4155V, 4158V, 4161V, 4164V, 4167V, 4170V, 4173V, 4176V, 4179V, 4182V, 4185V, 4188V, 4191V, 4194V, 4197V, 4200V, 4203V, 4206V, 4209V, 4212V, 4215V, 4218V, 4221V, 4224V, 4227V, 4230V, 4233V, 4236V, 4239V, 4242V, 4245V, 4248V, 4251V, 4254V, 4257V, 4260V, 4263V, 4266V, 4269V, 4272V, 4275V, 4278V, 4281V, 4284V, 4287V, 4290V, 4293V, 4296V, 4299V, 4302V, 4305V, 4308V, 4311V, 4314V, 4317V, 4320V, 4323V, 4326V, 4329V, 4332V, 4335V, 4338V, 4341V, 4344V, 4347V, 4350V, 4353V, 4356V, 4359V, 4362V, 4365V, 4368V, 4371V, 4374V, 4377V, 4380V, 4383V, 4386V, 4389V, 4392V, 4395V, 4398V, 4401V, 4404V, 4407V, 4410V, 4413V, 4416V, 4419V, 4422V, 4425V, 4428V, 4431V, 4434V, 4437V, 4440V, 4443V, 4446V, 4449V, 4452V, 4455V, 4458V, 4461V, 4464V, 4467V, 4470V, 4473V, 4476V, 4479V, 4482V, 4485V, 4488V, 4491V, 4494V, 4497V, 4500V, 4503V, 4506V, 4509V, 4512V, 4515V, 4518V, 4521V, 4524V, 4527V, 4530V, 4533V, 4536V, 4539V, 4542V, 4545V, 4548V, 4551V, 4554V, 4557V, 4560V, 4563V, 4566V, 4569V, 4572V, 4575V, 4578V, 4581V, 4584V, 4587V, 4590V, 4593V, 4596V, 4599V, 4602V, 4605V, 4608V, 4611V, 4614V, 4617V, 4620V, 4623V, 4626V, 4629V, 4632V, 4635V, 4638V, 4641V, 4644V, 4647V, 4650V, 4653V, 4656V, 4659V, 4662V, 4665V, 4668V, 4671V, 4674V, 4677V, 4680V, 4683V, 4686V, 4689V, 4692V, 4695V, 4698V, 4701V, 4704V, 4707V, 4710V, 4713V, 4716V, 4719V, 4722V, 4725V, 4728V, 4731V, 4734V, 4737V, 4740V, 4743V, 4746V, 4749V, 4752V, 4755V, 4758V, 4761V, 4764V, 4767V, 4770V, 4773V, 4776V, 4779V, 4782V, 4785V, 4788V, 4791V, 4794V, 4797V, 4800V, 4803V, 4806V, 4809V, 4812V, 4815V, 4818V, 4821V, 4824V, 4827V, 4830V, 4833V, 4836V, 4839V, 4842V, 4845V, 4848V, 4851V, 4854V, 4857V, 4860V, 4863V, 4866V, 4869V, 4872V, 4875V, 4878V, 4881V, 4884V, 4887V, 4890V, 4893V, 4896V, 4899V, 4902V, 4905V, 4908V, 4911V, 4914V, 4917V, 4920V, 4923V, 4926V, 4929V, 4932V, 4935V, 4938V, 4941V, 4944V, 4947V, 4950V, 4953V, 4956V, 4959V, 4962V, 4965V, 4968V, 4971V, 4974V, 4977V, 4980V, 4983V, 4986V, 4989V, 4992V, 4995V, 4998V, 5001V, 5004V, 5007V, 5010V, 5013V, 5016V, 5019V, 5022V, 5025V, 5028V, 5031V, 5034V, 5037V, 5040V, 5043V, 5046V, 5049V, 5052V, 5055V, 5058V, 5061V, 5064V, 5067V, 5070V, 5073V, 5076V, 5079V, 5082V, 5085V, 5088V, 5091V, 5094V, 5097V, 5100V, 5103V, 5106V, 5109V, 5112V, 5115V, 5118V, 5121V, 5124V, 5127V, 5130V, 5133V, 5136V, 5139V, 5142V, 5145V, 5148V, 5151V, 5154V, 5157V, 5160V, 5163V, 5166V, 5169V, 5172V, 5175V, 5178V, 5181V, 5184V, 5187V, 5190V, 5193V, 5196V, 5199V, 5202V, 5205V, 5208V, 5211V, 5214V, 5217V, 5220V, 5223V, 5226V, 5229V, 5232V, 5235V, 5238V, 5241V, 5244V, 5247V, 5250V, 5253V, 5256V, 5259V, 5262V, 5265V, 5268V, 5271V, 5274V, 5277V, 5280V, 5283V, 5286V, 5289V, 5292V,

Named left control. The shaft sticks on DIX steel and across the Mercury

Part 1/2: Type-L; Parts 3/4/5: Type-K SuperSpeed; Part 6: Type-A USB 3.0.
Configuration 3 per 06000038009. Schemas 3-4/3-5: HS30P and HSUSL247 driven by PD HCU, not High-voltage HSUS.



One 100 pf capacitor per V0033/V0380 pin, placed at the relevant pin. Bulk capacitors near hub.
220 pf series capacitors on each hub SuperSpeed TX line only. Rcvr 90 ohm differential pairs without Klabs.

